

Brands

PERSONAL

Transcend. Transcend's JetFlash T5 USB flash drive has recently won the 'G-Mark' distinction for its design at the 2009 Good Design Award competition. It is available in 2GB, 4GB and 8GB capacities.

Toshiba. The company has come up recently with its 4.6cm (1.8-inch) Serial ATA HDD product line.

Kingston. Kingston's latest offering includes the SSDNow V Series 40GB Boot Drive that boasts to accelerate a desktop's boot, shutdown and application load times.

Western Digital. Western Digital has added 2 TB Green hard drive to its kitty.

SanDisk. SanDisk has come up with what it calls its fastest 32GB SDHC card in India.

ConJoin. ConJoin has unveiled its automated backup device named Clickfree.

ENTERPRISE

IBM. IBM has put its footprints into the cloud storage market with the IBM Smart Business Storage Cloud solution.

EMC. EMC has rolled out its Symmetrix V-Max virtual datacentre recently companies.

HP. HP has its new shared storage solutions—HP LeftHand P400 SAN and HP StorageWorks SB40c—in the market.

Hitachi. Of late, Hitachi has launched two new hard drives for enterprise storage application—the Ultrastar C15K147 and 600GB Ultrastar 15K600.

Buffalo. Buffalo has recently served its Drive Station CX series to the enterprise market in India.

NEC. NEC has recently launched NEC Storage E-Series that integrates iSCSI interface or Fibre channel interface.

Symantec. Symantec has started delivering scalable cloud storage platform—FileStore—to companies.

new. Plus, firms also offer specialised data migration services itself.

NEC's HYDRAStor is also equipped for this transition, says Abhilesh Guleria, country head-IT/ITP, NEC India. "NEC's design direction has been to design a unique 'community of smart nodes' enabling IT to non-disruptively ride both technology and growth curves," he adds.

Joshi of Symantec feels that search, discovery and classification is another area in the storage software space where drastic changes would be seen, especially with big enterprises which comprise lots of servers and unstructured data.

Green storage

Green storage has been battered to death in the past year. Energy-efficient storage devices, thin provisioning, deduplication, virtualisation and other technologies to optimise storage utilisation and thereby reduce power-consumption; environment-friendly production and disposal; RoHS and other regulatory compliances, and much more.

Interesting technologies continue to emerge in this connection. "Last year, we have seen a disk-level approach taking hold when devising strategies to reduce energy consumption. EMC implemented disk spin-down, which reduces the amount of power consumed, as the disk slips into a sleep mode after the backup is completed," comments Gupta.

New technologies

Racetrack technology. "Racetrack technology is going to redefine storage," feels Sivaswamy. "This recent invention is a new form of digital storage technology that delivers manifold increases in both storage capacity and battery life of handheld gadgets like MP3 and MP4 players, mobile phones, etc," he adds.

Low-energy lasers. P.K. Gupta thinks that the usage of low-energy lasers to grow crystals as a storage medium in salt solutions will become popular. It has the potential to hold 100 times more data than current systems. Holographic and molecular methods for storage are also underway.

Nano-structured systems. "Re-

cently, engineers in the NC State University, USA, have created a new fingernail-sized chip that can hold 1 terabyte of data—50 times the capacity of today's best silicon-based chip technologies, using a nano-structured Ni-MgO system. In real terms, it can store up to 20 HD-DVDs or 250 million pages of text," says Gupta.

Storage class memory. Next in line is Storage Class Memory (SCM). SCM is an SSM that blurs the boundaries between storage and memory by being low-cost, fast and non-volatile.

Digital Rosetta Stone. Another futuristic technology worth noting is the 'Digital Rosetta Stone' developed by Tadahiro Kuroda, an electrical engineering professor at Keio University in Japan. This wireless storage device comprises four thin wafers, each 38cm in diameter, embedded with 1100 tiny chips. These four wafers will have 312 GB of memory, the equivalent of 480 CDs. The device needs no wires, and allows information to be beamed wirelessly on radio waves.

Small and bright holograms. General Electric's breakthrough is a development that could bring holographic storage to the mass market. The media developed by GE relies on smaller, less complex holograms. These holograms have shown a 200-fold increase in reflective power compared to the earlier one.

"Individuals are creating 70 per cent of the world's digital data and need devices to help them store and protect this information," says Gupta. "Gadgets such as digital cameras, digital TV shipments, surveillance cameras, sensor-based applications, mobiles, etc, are contributing to this growth," he adds.

Given that the demand for storage will continue to increase at both the personal and enterprise realms, there is no betting against the fact that storage will continue to see more innovations. ●

The author is a freelance writer based in Bengaluru. She writes on a variety of topics, her favourites being technology, cuisine, and life